

Linear model selection methods

Linear model selection methods

- ▶ $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$
- ▶ Conflicting goals: prediction accuracy vs model interpretability
- ▶ What if p is quite large in relation to n ?
 - ▶ We have, potentially, a lot of information about the response
 $\implies$ Good predictions
 - ▶ We have to estimate many parameters $\implies$ Estimation uncertainty $\implies$ Bad predictions
 - ▶ A trade-off between prediction bias (due to an incorrectly specified model) and prediction variance (due to too many parameters to estimate)

Three different type of techniques

- ▶ Subset selection: Find the set of predictors that gives the best predictions.
- ▶ Shrinkage: Use all predictors but restrict the parameter space.
- ▶ Dimension reduction: Find combinations of predictors.

Best subset selection

1. $\mathcal{M}_0$ is the model with no predictors, $Y = \beta_0 + \varepsilon$
 2. For $k = 1, 2, \dots, p$
 - a. Fit all $\binom{p}{k}$ models with k predictors
 - b. Pick the best model among them, e.g. the one with largest R^2 or, equivalently, minimize RSS (here we can do this because all models have the same number of parameters). Call this model $\mathcal{M}_k$
 3. Select the best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, e.g. with largest *adjusted R^2 or smallest testMSE (here we cannot use the raw R^2 since it does not control for overfitting)*.
- 2^p models. If $p = 50$ we would have to estimate more than 10^{15} models.

Example with Auto data (miles per gallon)

From the R-help for the data (help(Auto))

Number	Name
Dependent	mpg
1	cylinders
2	displacement
3	horsepower
4	weight
5	acceleration
6	age

Example with Auto data (miles per gallon)

```
# ALL PREDICTORS
require(ISLR)
Auto$age=83-Auto$year
Auto=Auto[,!(names(Auto) %in% c("name","origin","year"))]
regall=lm(mpg~.,data=Auto)
summary(regall)

##
## Call:
## lm(formula = mpg ~ ., data = Auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.6927 -2.3864 -0.0801  2.0291 14.3607
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  47.9942254   2.1625355   22.193  <2e-16 ***
## cylinders    -0.3298591   0.3321041   -0.993   0.321
## displacement  0.0076784   0.0073577    1.044   0.297
## horsepower   -0.0003914   0.0138365   -0.028   0.977
## weight       -0.0067946   0.0006700  -10.141  <2e-16 ***
## acceleration  0.0852732   0.1020356    0.836   0.404
## age          -0.7533672   0.0526181  -14.318  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.435 on 385 degrees of freedom
## Multiple R-squared:  0.8093, Adjusted R-squared:  0.8063
## F-statistic: 272.2 on 6 and 385 DF, p-value: < 2.2e-16
```

Step 1 in R - Fit $\mathcal{M}_0$

Run a regression without any variables!!!

```
y=Auto$mpg  
M0=lm(y~1)  
M0
```

```
##  
## Call:  
## lm(formula = y ~ 1)  
##  
## Coefficients:  
## (Intercept)  
##      23.45
```

```
mean(y)
```

```
## [1] 23.44592
```

```
RSS=sum(residuals(M0)^2)  
RSS
```

```
## [1] 23818.99
```

Step 2 in R

- Find best model of size $k = 1, 2, \dots, p$, respectively.
- Example for $k=2$

Variables		RSS
1	2	8342.57
1	3	8172.49
1	4	7206.11
1	5	9385.44
1	6	6786.73
2	3	7995.34
2	4	7170.31
2	5	8371.71
2	6	6172.62
3	4	6993.85
3	5	8804.96
3	6	7491.14
4	5	7152.89
4	6	4568.95
5	6	14102.75

- Among the models with 2 predictors, the model with variables number 4 (weight) and 6 (age) is the one with lowest RSS, or equivalently, highest R^2 .

Step 3 in R

- ▶ We showed that $\mathcal{M}_2$ is the model $Y = \beta_0 + \beta_4 X_4 + \beta_6 X_6 + \varepsilon$
- ▶ Similarly we can find $\mathcal{M}_k$ for all other $k = 1, \dots, p$

	cylinders	displacement	horsepower	weight	acceleration	age	adjR2
	0	0	0	0	0	0	0.0000
	1	0	0	0	1	0	0.6918
▶	2	0	0	0	1	0	0.8072
	3	0	0	0	1	1	0.8071
	4	0	1	0	1	1	0.8068
	5	1	1	0	1	1	0.8068
	6	1	1	1	1	1	0.8063

- ▶ Result: Use weight and age
- ▶ We have estimated 64 models

Use built-in function

```
require(leaps)
```

```
## Loading required package: leaps
```

```
reg2=regsubsets(mpg~.,data=Auto,nvmax = 6)
```

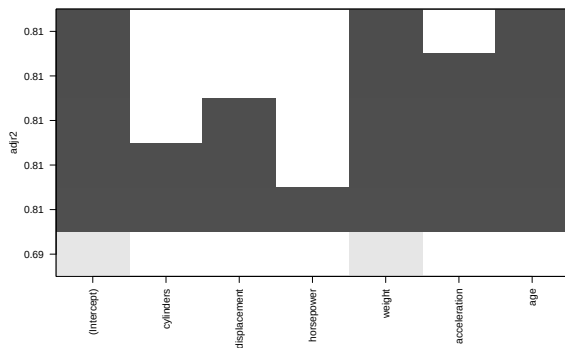
```
reg3sum=summary(reg2)
```

```
cbind(reg3sum$which[,-1],adjR2=round(reg3sum$adjr2,4))
```

##	cylinders	displacement	horsepower	weight	acceleration	age	adjR2
## 1	0	0	0	1	0	0	0.6918
## 2	0	0	0	1	0	1	0.8072
## 3	0	0	0	1	1	1	0.8071
## 4	0	1	0	1	1	1	0.8068
## 5	1	1	0	1	1	1	0.8068
## 6	1	1	1	1	1	1	0.8063

Another presentation of the result

```
plot(reg2,scale="adjr2",col=gray.colors(1000))
```



Result: Use weight and age

Forward stepwise selection

- ▶ A problem with best subset selection is that we have to estimate all possible models. Sometimes not feasible.
- 1. $\mathcal{M}_0$ is the model with no predictors, $Y = \beta_0 + \varepsilon$
- 2. For $k = 0, 1, \dots, p - 1$
 - a. Fit all $p - k$ models that adds one single predictor to $\mathcal{M}_k$
 - b. Pick the best model among them, e.g. the one with largest R^2 .
Call this model $\mathcal{M}_{k+1}$
- 3. Select the best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, e.g. with largest *adjusted R^2* or *smallest testMSE*
- ▶ We have to estimate $\frac{p(p+1)}{2} + 1$ models. $p = 50$ implies 1276 models

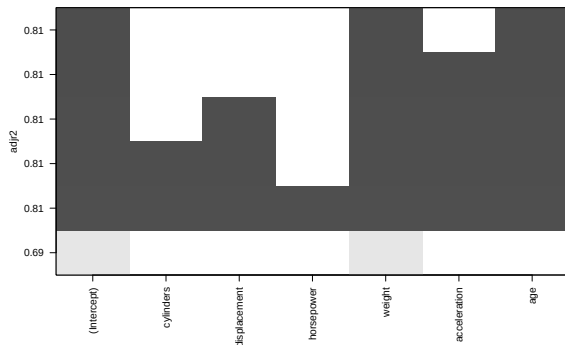
Forward stepwise selection for the Auto data

```
reg3=regsubsets(mpg~.,data=Auto,nvmax = 6,method="forward")
reg3sum=summary(reg3)
cbind(reg3sum$which[,-1],adjR2=round(reg3sum$adjr2,4))
```

##	cylinders	displacement	horsepower	weight	acceleration	age	adjR2
## 1	0	0	0	1	0	0	0.6918
## 2	0	0	0	1	0	1	0.8072
## 3	0	0	0	1	1	1	0.8071
## 4	0	1	0	1	1	1	0.8068
## 5	1	1	0	1	1	1	0.8068
## 6	1	1	1	1	1	1	0.8063

Another presentation of the result

```
plot(reg3,scale="adjr2",col=gray.colors(1000))
```



Result: Use weight and age

We have estimated 21 models

Backward stepwise selection

- ▶ A problem with best subset selection is that we have to estimate all possible models. Sometimes not feasible.
- 1. $\mathcal{M}_p$ is the model with all predictors, $Y = \beta_0 + \sum_{k=1}^p \beta_k X_k + \varepsilon$
- 2. For $k = p, p-1, \dots, 1$
 - a. Fit all k models with all but one of the predictors in $\mathcal{M}_k$
 - b. Pick the best model among them, e.g. the one with largest R^2 .
Call this model $\mathcal{M}_{k-1}$
- 3. Select the best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, e.g. with largest *adjusted R^2* or *smallest testMSE*
- ▶ We have to estimate $\frac{p(p+1)}{2} + 1$ models. $p = 50$ implies 1276 models

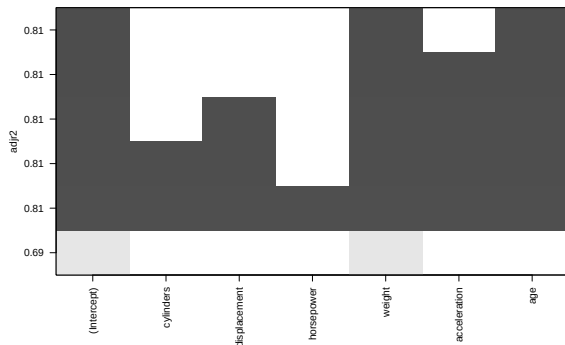
Backward stepwise selection for the Auto data

```
reg4=regsubsets(mpg~.,data=Auto,nvmax = 6,method="backward")
reg4sum=summary(reg4)
cbind(reg4sum$which[,-1],adjR2=round(reg4sum$adjr2,4))
```

##	cylinders	displacement	horsepower	weight	acceleration	age	adjR2
## 1	0	0	0	1	0	0	0.6918
## 2	0	0	0	1	0	1	0.8072
## 3	0	0	0	1	1	1	0.8071
## 4	0	1	0	1	1	1	0.8068
## 5	1	1	0	1	1	1	0.8068
## 6	1	1	1	1	1	1	0.8063

Another presentation of the result

```
plot(reg4,scale="adjr2",col=gray.colors(1000))
```



Result: Use weight and age

We have estimated 21 models

An example with 3 x-variables

Predict Y based on a linear function of X_1 , X_2 and X_3 .

We shall work through the examples for

- ▶ Best subset selection
- ▶ Forward stepwise selection
- ▶ Backward stepwise selection

Best subset selection

1. $\mathcal{M}_0 : Y = \beta_0 + \varepsilon$

2. For $k = 1, 2, 3$

▶ $k = 1$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_1 X_1 + \varepsilon \\ Y = \beta_0 + \beta_2 X_2 + \varepsilon \\ Y = \beta_0 + \beta_3 X_3 + \varepsilon \end{array} \right\} \quad \begin{array}{l} \text{Pick the model} \\ \text{with highest } R^2. \\ \text{Call it } \mathcal{M}_1. \end{array}$$

▶ $k = 2$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon \\ Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \varepsilon \\ Y = \beta_0 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon \end{array} \right\} \quad \begin{array}{l} \text{Pick the model} \\ \text{with highest } R^2. \\ \text{Call it } \mathcal{M}_2. \end{array}$$

▶ $k = 3$

$$\mathcal{M}_3 : Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

3. Choose the model from $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2$ and $\mathcal{M}_3$ with largest adjusted R^2 , smallest testMSE or...

Best subset selection

- ▶ We have estimated $2^3 = 8$ models. (-)
- ▶ We have estimated all possible models (+), compared their fit to data and chosen the one with highest adjusted R^2 .
- ▶ Adjusted R^2 can be substituted with other measures of fit. The important property of such a measure is that it controls for overfitting. This means that if we compare two models with the same value on the measure, the model with fewer parameters should be chosen.

Measures that accounts for overfitting

Building blocks: $TSS = \sum (y_i - \bar{y})^2$ and $RSS = \sum (y_i - \hat{y}_i)^2$.
 d = number of predictors, $\sigma^2 = \text{Var}(\varepsilon)$.



$$\text{Adjusted } R^2 = 1 - \frac{RSS/(n - d - 1)}{TSS/(n - 1)}$$

- ▶ Mallow's C_p :

$$C_p = \frac{1}{n} (RSS + 2d\hat{\sigma}^2)$$

- ▶ Akaike information criteria

$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS + 2d\hat{\sigma}^2)$$

- ▶ Bayesian information criteria

$$BIC = \frac{1}{n} (RSS + \log(n)d\hat{\sigma}^2)$$

Forward stepwise selection

1. $\mathcal{M}_0: Y = \beta_0 + \varepsilon$

2. For $k = 0, 1, 2$

► $k = 0$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_1 X_1 + \varepsilon \\ Y = \beta_0 + \beta_2 X_2 + \varepsilon : \mathcal{M}_1 \\ Y = \beta_0 + \beta_3 X_3 + \varepsilon \end{array} \right\} \quad \text{In this example we assume that the second model had the highest } R^2.$$

► $k = 1$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_2 X_2 + \beta_1 X_1 + \varepsilon \\ Y = \beta_0 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon : \mathcal{M}_2 \end{array} \right\} \quad \text{The second model had the highest } R^2.$$

► $k = 3$

$$\mathcal{M}_3 : Y = \beta_0 + \beta_2 X_2 + \beta_3 X_3 + \beta_1 X_1$$

3. Choose the model from $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2$ and $\mathcal{M}_3$ with largest adjusted R^2 , smallest testMSE or...

Forward stepwise selection

- ▶ We have estimated $\frac{4 \cdot 3}{2} + 1 = 7$ models. (+)
- ▶ We have not estimated all possible models (-), compared their fit to data and chosen the one with highest adjusted R^2 .
- ▶ Again Adjusted R^2 can be substituted with other measures of fit.

Backward stepwise selection

1. $\mathcal{M}_3 : Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$

2. For $k = 3, 2, 1$

► $k = 3$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon \\ Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \varepsilon : \mathcal{M}_2 \\ Y = \beta_0 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon \end{array} \right\} \quad \text{In this example we assume that the second model had the highest } R^2.$$

► $k = 2$

$$\left. \begin{array}{l} Y = \beta_0 + \beta_1 X_1 + \varepsilon : \mathcal{M}_1 \\ Y = \beta_0 + \beta_3 X_3 + \varepsilon \end{array} \right\} \quad \text{The second model had the highest } R^2.$$

► $k = 1$

$$\mathcal{M}_0 : Y = \beta_0 + \varepsilon$$

3. Choose the model from $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2$ and $\mathcal{M}_3$ with largest adjusted R^2 , smallest testMSE or...